



## Wout Deleu

Java Consultant

Gent, Belgium

13/12/2001

I'm a software engineer with a strong preference for backend development, drawn to the architecture, logic, and precision that goes into building robust systems. Through my work at Volvo Cars on supply chain software with the DEVEX team, I've developed an appreciation for clean, well-structured code and scalable solutions. My analytical mindset and drive for structured problem-solving are constants, whether I'm designing a service, debugging a pipeline, exploring a new technology, or looking for ways to improve how a team works. During my studies I also reached the finals of the CyberSecurity Challenge Belgium, further sharpening these instincts.

I continuously invest in broadening my technical knowledge through certifications and personal projects.

Outside of work, I enjoy getting my hands dirty with Linux systems, self-hosted servers, and personal infrastructure, finding useful and fun ways to enhance my home or workflows. Setting up dashboards, writing scripts, building out my homelab, exploring tools for cleaner development environments... there's a particular satisfaction in making a system work exactly the way you want it to.

Sport is a big part of my life. I play badminton at a Flemish competitive level, ranked as high as 60th in Belgium and finishing in the top 16 at this year's Belgian Championships, and hold a Trainer B diploma. I coach three individual top players, assist the West-Flemish talent team Open Elite, and lead weekly group sessions at De Pluimplukkers, the largest badminton club in Belgium. I also keep active through running, swimming, and calisthenics, and have completed a few half marathons along the way.

When I'm not training or coaching, I like to unwind with a good fantasy novel, a movie, or a series. You'll also find me hunting for a great coffee, planning where to travel next, or tinkering with something on a server.

## Skills

|             |                                                         |                  |                                                         |
|-------------|---------------------------------------------------------|------------------|---------------------------------------------------------|
| Java        | <div><div></div><div></div><div></div><div></div></div> | Spring Framework | <div><div></div><div></div><div></div><div></div></div> |
| Python      | <div><div></div><div></div><div></div><div></div></div> | C++              | <div><div></div><div></div><div></div><div></div></div> |
| Rust        | <div><div></div><div></div><div></div><div></div></div> | C#               | <div><div></div><div></div><div></div><div></div></div> |
| Angular     | <div><div></div><div></div><div></div><div></div></div> | React            | <div><div></div><div></div><div></div><div></div></div> |
| Arduino     | <div><div></div><div></div><div></div><div></div></div> | SQL              | <div><div></div><div></div><div></div><div></div></div> |
| NoSQL       | <div><div></div><div></div><div></div><div></div></div> | MongoDB          | <div><div></div><div></div><div></div><div></div></div> |
| Linux       | <div><div></div><div></div><div></div><div></div></div> | AWS              | <div><div></div><div></div><div></div><div></div></div> |
| Docker      | <div><div></div><div></div><div></div><div></div></div> | Pipelines        | <div><div></div><div></div><div></div><div></div></div> |
| Agile/Scrum | <div><div></div><div></div><div></div><div></div></div> | Excel            | <div><div></div><div></div><div></div><div></div></div> |
| Keycloak    | <div><div></div><div></div><div></div><div></div></div> | Idap connections | <div><div></div><div></div><div></div><div></div></div> |

Kafka  
Project  
Management



Oracle db  
Analysis



## Projects



### Volvo Cars Belgium

Software Developer

05/2024 - Present

As a part of the Developer Experience Team, we develop software utilized by other teams, creating common tools and components primarily using Java with the Spring Framework and Angular. However, my role extends beyond coding common components. We are also responsible for setting development standards, assisting with Kafka-related issues, maintaining and migrating specific codebases, and even handling DevOps tasks such as managing pipelines and writing bash scripts. This multifaceted position allows me to contribute across various areas, ensuring smooth and efficient development processes within the organisation.

The team was newly formed within the organization, operating without a PM, PO, or Scrum Master. Recognizing the need for structure, I took the initiative to enhance our workflow—introducing retrospectives, improving team organization, and fostering a more efficient and collaborative development process.

One of my key focus areas was developing a Keycloak Integration Tool, enabling management to seamlessly evaluate, configure, and oversee the authorization system across all Volvo plants. This was a project I led from the ground up—starting with gathering requirements, designing the architecture, and driving both front-end and back-end development. By streamlining access management, this tool significantly improved efficiency and security across the organization.

I served also as the sole developer, analyst, and architect of a production microservice bridging Maximo (the master WMS) and S4M vending machines on the factory floor. I designed the full solution end-to-end - from domain analysis and architectural decisions to implementation and testing. The service handles bidirectional Kafka event streaming, transforming and repartitioning messages to keep stock levels and transactions in sync across both platforms. Built with Java, Spring Boot, and hexagonal (ports & adapters) architecture, the project became a leading internal example of software quality - recognized for its design patterns, test architecture, thorough documentation, and production-grade observability.

### Tasks

- Analyzing, designign and developping a Keycloak integration tool for business users.
- Setting up a Kafka Metric system using Elastic and CosmosDB to evaluate and collect kafka metrics in the different plants!
- Analyzing, gattering requirements and designing a Microservice to configure a Vending Machine of an external party on the factory floor.

### Skills

Spring Framework   spring security   Spring Boot   Keycloak   KeyCloak   DevOps  
 Docker Compose   Docker/Kubernetes   Kustomize   Domain Driven Design  
 Hexagonal architecture   Active Directory   Scrummaster   Angular



## Silverfin

Solution Developer

02/2024 - 05/2024

As a consultant, I collaborated within a team of *17 professionals*, contributing to the development of templates utilizing a variant of the **Liquid** template language, specifically designed for Silverfin. My primary focus involved the creation of templates tailored to manage Personal Income Tax for accountants. Faced with a highly time-sensitive project, I was added to the team in order to provide extra help and try to achieve the successful completion of tasks within the prescribed urgent deadline.

The project itself was a little outside of my comfort zone, but I feel this is only an upside in my personal journey as a developer!

### Tasks

- Developing templates in the Silverfin language (Liquid variant)
- Code reviews

### Skills

Liquid Scrum JIRA GitHub Teamwork Excel



## Axxes

Frontend Developer

12/2023 - 01/2024

As a Junior Developer at AXXES, I actively participated in the development of the **Carpool Tool**, an internal application designed to streamline the management of carpools. The tool facilitates user subscription to events and allows them to specify their carpool preferences. Additionally, the system autonomously generates carpool proposals, which can be further adjusted by staff to accommodate changing preferences or unexpected irregularities through an algorithmic approach.

Despite my inclination towards backend development, I intentionally sought the opportunity to strengthen my frontend skills by working on the Carpool Tool. This experience provided valuable insights into building real-world applications and enhancing my proficiency in collaborative teamwork.

Throughout the project's evolution, the team underwent dynamic changes as members transitioned to different projects. I initially served as the sole frontend developer, collaborating closely with a backend counterpart. As the project progressed, a second frontend developer joined our efforts, fostering a cohesive team dynamic that resulted in significant advancements and achievements.

This project not only honed my programming skills but also afforded me the opportunity to experience working in a team, as well as controlling and managing the project.

### Tasks

- Managing the Scrum Board
- Figma Designs and Flow charts of the site
- Designing the algorithm (theoretically) of the carpool generation
- React development

### Skills

Ownership / management of a project. React Scrum JIRA Azure Algorithms  
Figma

## KU Leuven

Thesis Student

09/2022 - 06/2023

"Time Is Running Out Assessing Temporal Privacy of Privacy Zones in Fitness Tracking Social Networks" - Reconstructing routes cut off by privacy zones from fitness apps.

Fitness tracking social networks such as Strava allow users to connect with each other by sharing sports activities. By sharing these activities, users are unintentionally revealing sensitive location data, which allows **malicious users to infer their home address**. This is why these platforms introduce functionality allowing users to hide portions of the trajectory of the activity, in order to protect sensitive locations. Previous research has shown that in spite of these countermeasures, they are still vulnerable to attacks. These inference attacks can use either distance data or elevation data to discover sensitive locations.

During my thesis, I wrote **algorithms** that could **reconstructs cut off parts of the trajectory, exposing home locations of users**. I also **investigated in which circumstances this was most effective**, and finally what countermeasures could be applied to prevent this from happening.

*Team: Individual, under supervision*

### Tasks

- Literature Study of fitness tracking social networks and their privacy
- Literature Study of previous attack models
- Implementation of an attack model
- Writing a scientific article

### Skills

Python Data visualization Analysis Presenting/Public speaking Algorithms

## KU Leuven

Student

09/2022 - 12/2022

Developing an app which is able to track all the expenses made in a period of time, targeted to everyday users. You can also find some financial news articles, helping you to stay on top of your investments.

*Team: 2 students*

### Tasks

### Skills

Flutter Dart



## PSI Control

Student Job

07/2022 - 09/2022

R&D employee designing a small adjustable power supply using Arduino, electronic circuit design and C#, powered by USB-C.

I extracted the power supply functionality out of an existing mother board, and redesigning that part into a portable form factor. Then, I programmed an **Arduino (C++)** to control the power supply, using a GUI. I made sure the **SOLID** principles were taken into account, by building multiple abstraction layers, e.g. the controller of the physical pins on the Arduino was layer one, the last layer was the GUI. The functionality of the program can be easily expanded thanks to the abstractions.

A first real work experience, learning to work in a large corporation. Turning **requirements** into real life **solutions**, as well as following the standards of a company really helped in my evolution as a developer.

*Team: Individual with possibility of feedback.*

### Tasks

- Isolating functionality out of a functioning power source
- Translating designed power source to its functionality
- Designing C-program to control the arduino, in a layered fashion
- Designing a GUI in C# to control the arduino

### Skills

Arduino

C++

C#

Electronics

WPF

## KU Leuven

Student

09/2021 - 06/2022

Bachelor project. Designing a full stack app (Website + Backend + Mobile App), which enables users to select and assign Master Theses. Users are able to log in with different roles and different functionality. Coordinators can add, modify and assign master theses to specific students. Students can see and mark different theses as preferred, and rank them for themselves.

First time developing a platform using a dedicated backend and frontend. Also the first contact with frameworks (**React**), as well as **Java Spring**. Also working coordinated in a team, to deliver the best possible product possible in the limited amount of the time.

*Team: 3 students (each had his own role: backend, frontend, **both**)*

### Tasks

- Designing architecture of the project
- Designing database structure
- Writing backend controllers, services and repositories
- Creating the mobile app

### Skills

React   React Native   Java   Spring Framework   JavaScript



## KU Leuven

Student

02/2021 - 12/2022

Across different courses and periods, the subject of a port where ships arrive and depart, as well as load and unload containers. The yard itself is managed by cranes, working simultaneously. During these projects, we handled different aspects of the port. We wrote **deterministic algorithms** to plan the crane tasks and movements. We scheduled different processes on the port, using non-deterministic **AI-algorithms**. Finally, we brought everything together in a **full simulation model with a visualization**. This makes it possible to get characteristics and stats of different possible scenarios, as well as detect possible problems with hypothetical cases.

This was a large project that had a lot of algorithmic and logical thinking, in which I therefore made great progress. Also the size of the project required careful planning, and good arrangements to deliver parts which can work together!

*Team: 3 students*

### Tasks

- Emulate crane movement
- Implementing Container Yard ordering
- Implementing the AI process scheduler
- Analyzing and visualizing real world data, and calculating statistical formulas used for the simulation
- Visualization

### Skills

Python Python scripting Java Algorithms Data science Data visualization

## Education



**KU Leuven - Industrieel Ingenieur Elektronica ICT**

Master

09/2019 - 06/2023

## Certificates



**Spring Professional Develop (2V0-72.22)**

achieved: 08/2025



**PSM I**

achieved: 02/2025



**OCP Java SE17**

achieved: 01/2024

## Trainings



**Axxes Traineeship**

16/08/2023 - 22/09/2023

**SOFTWARE PRINCIPLES:** *Clean Code, SOLID, Mocking & Unit Testing, Agile*

**BACKEND:** *Spring Intro, Spring Data, Spring Security*

**FRONTEND:** *JavaScript, HTML/CSS, React, Angular*

**MICROSERVICE ARCHITECTURE:** *REST, Microservices, Event-Driven, API Testing*

**CLOUD:** *Docker, Linux, AWS, MongoDB, CI/CD, Microservices, Event-Driven*

**DEEP DIVES:** *Modern Java, Data, Spring Security, Dealing with legacy, Automated-Front-End-Testing*

## Languages

|         | Understanding | Speaking | Writing      |
|---------|---------------|----------|--------------|
| Dutch   | Native        | Native   | Native       |
| English | Fluent        | Fluent   | Fluent       |
| French  | Fluent        | Fluent   | Intermediate |